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Analysis of new top-hat transformation and the application for infrared dim small target detection

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ABSTRACT

To improve the performance of top-hat transformation for infrared dim small target detection in a simple and effective way, the definition, properties, multi-scale operations of new top-hat transformation and the application for infrared dim small target detection are addressed in this paper. The definition of new top-hat transformation uses two different but correlated structuring elements to reorganize the classical top-hat transformation, and takes into account of the difference information between the target and surrounding regions. Given this definition, the new top-hat transformation has some special properties and three types of multi-scale operations, which are discussed in detail. Subsequently, one application case of multi-scale operation for noise suppression is given. Good performance of the application for infrared dim small target detection is obtained, which could be ascribed to the proper selection of structuring elements based on the properties. The experimental results of the application demonstrate that new top-hat transformation can detect infrared dim small target more efficiently than classical top-hat transformation and some other widely used methods.

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1. Introduction

Mathematical morphology was firstly proposed and extended for image analysis by Matheron and Serra [1]. Mathematical morphology is based on geometry and set theory. It has been well used and developed in different applications [2–9]. Small target detection is one of the important applications of mathematical morphology, and the top-hat transformation is an important operation in this application [7–10].

Although the top-hat transformation has been used for small target detection, the ability of target detection is weak. When the signal to noise ratio (SNR) of image is high, top-hat transformation can be directly used for small target detection [7]. But, when SNR is low, the target is dim, and the noises and clutter are heavy, which heavily affect the performance of top-hat transformation. In this situation, top-hat transformation cannot detect small target well, and may even decrease the SNR of original image and cause target losing. To enhance the adaptability of top-hat transformation to noises, heavy clutter and dim target intensity, and thus to improve the performance of top-hat transformation for small target detection, many improved algorithms are proposed [8–10]. Increasing SNR of image through energy cumulating before target detection can suppress the effect of noises and heavy clutter [8], but this algorithm does not solve the problem of top-hat transformation. Structuring element is one of the crucial parts of top-hat transformation, and is also the crucial part of the performance improvement of top-hat transformation. To find an appropriate structuring element for target detection, algorithms based on neural network and genetic algorithm are proposed [9]. Although these algorithms perform well in some cases, they do not directly improve the top-hat transformation. Constructing new top-hat transformation through changing the structuring elements to suppress the effect of noises is a direct way to improve the performance of top-hat transformation [10]. However, it has not been used for infrared small target detection as far as we know. A new top-hat transformation has been defined and well used in our previous work [11]. In this paper, the definition, properties, multi-scale operations of new top-hat transformation and its application for infrared dim small target detection are discussed in detail. Comparing with the new top-hat transformation, the original top-hat transformation is called classical top-hat transformation. Firstly, some shortcomings of classical top-hat transformation are given. Then, the definition of new top-hat transformation is given through importing two different but correlated structuring elements into top-hat transformation and reorganizing classical top-hat transformation. This definition is based on suppressing the shortcomings of classical top-hat transformation. The properties and three types of multi-scale operations of new top-hat transformation are analyzed, which shows the superiority of new top-hat transformation.

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To ensure that the new top-hat transformation can be used well in the application of infrared small target detection, the selection of structuring elements is studied following the properties. Various experiments of the application of infrared dim small target detection verified that new top-hat transformation was better than classical top-hat transformation and other widely used methods.

This paper is organized as follows. Section 2 presents the definitions of classical top-hat transformation. Section 3 gives the shortcomings of classical top-hat transformation and defines new top-hat transformation. Section 4 studies the properties of new top-hat transformation. Section 5 discusses the multi-scale operations of new top-hat transformation. Section 6 demonstrates the application of new top-hat transformation for infrared small target detection. And, Section 7 is a conclusion.

2. Classical top-hat transformation

All the mathematical morphology operations are based on two basic operations: dilation and erosion. They work with two sets. One set is the original image to be analyzed and the other set is structuring element. Let f and B represent the gray-scale image and structuring element, respectively. The dilation and erosion of $f(x, y)$ by $B(u, v)$, denoted by $f \oplus B$ and $f \ominus B$, are defined by

$$(f \oplus B)(x,y) = \max_{u,v}(f(x-u, y-v) + B(u,v)),$$

$$(f \ominus B)(x,y) = \min_{u,v}(f(x+u, y+v) - B(u,v)),$$

where the domain of $f \oplus B$ and $f \ominus B$ are the dilation and erosion of the domain of f with the domain of B. Dilation (erosion) makes the image gray value larger (smaller) than that of the original image because of the maximum (minimum) operation. So, dilation (erosion) increases the size of bright (dark) region and decreases the size of dark (bright) region because of the maximum (minimum) operation.

If the structuring element is flat, that is

$$B(u,v) = 0, \quad \forall u, v,$$

dilation and erosion become

$$(f \oplus B)(x,y) = \max_{u,v}(f(x-u, y-v)),$$

$$(f \ominus B)(x,y) = \min_{u,v}(f(x+u, y+v)).$$

Based on dilation and erosion, opening and closing of $f(x, y)$ by $B(i, j)$, denoted by $f \circ B$ and $f \bullet B$, are defined by

$$(f \circ B)(x,y) = (f \ominus B) \oplus B,$$

$$(f \bullet B)(x,y) = (f \oplus B) \ominus B.$$

Opening smoothes bright small regions of image, and closing eliminates dark small holes.

Then, white top-hat transformation and black top-hat transformation of image f, denoted by WTH and BTH, are defined by

$$WTH(x,y) = f(x,y) - (f \circ B)(x,y),$$

$$BTH(x,y) = (f \bullet B)(x,y) - f(x,y).$$

WTH finds out bright regions in f, and BTH gives out dark regions. Small target region in image is usually a small bright (black) region. So, WTH (BTH) can be used directly to detect potential targets [7].

3. New top-hat transformation

3.1. Shortcomings of classical top-hat transformation

Structuring element is the crucial part of mathematical morphology operations. Small target is usually different from the surrounding background regions. So, there are difference information between the small target and the surrounding regions. The difference information is the reason of top-hat transformation being used for small target detection. So, top-hat transformation can detect target by using this difference information. But, classical top-hat transformation uses two same structuring elements in the operations of opening or closing, and does not utilize this difference information in the structuring element appropriately. This may decrease the efficiency of classical top-hat transformation to differentiate the real target region and noise or clutter regions. Therefore, the noise or clutter may affect the performance of classical top-hat transformation.

(1) The noises of image may lead to target losing or increase the number of false alarms when classical top-hat transformation is used for target detection.

The negative noise existed in target region leads to the target losing because of the two same structuring elements used in classical top-hat transformation [10]. Also, opening is usually used for positive noise filtering [1]. Then, the positive noise may be recognized as possible targets by classical top-hat transformation. So, the noises may affect the performance of classical top-hat transformation. These noises will be the potential false alarms.

(2) The classical top-hat transformation cannot differentiate the heavy clutter and real target region.

The definitions of classical top-hat transformations [1] indicate that all the pixels overlaid by the two same structuring elements are used for target recognition. Factually, most of the pixels locate in background. Then, if there are clutter backgrounds, most of the clutter will have outputs in the result image. So, the classical top-hat transformation cannot well differentiate the heavy clutter and target region.

Therefore, using two same structuring elements and thus inappropriately considering the differences between the target and surrounding regions result in the shortcomings of classical top-hat transformations. These shortcomings affect the performance of top-hat transformation for image processing.

3.2. Definitions

Target region in image is usually different from the surrounding background, based on which the top-hat transformation can be used for target detection. However, the shortcomings of classical top-hat transformation indicate that classical top-hat transformation cannot well utilize the difference information between the target region and surrounding background, which can be ascribed to the using of two same structuring elements. So, if the difference information between target region and surrounding background is directly well used in structuring element; the performance of top-hat transformation will be significantly improved. In this regards, a new top-hat transformation was proposed [11].

Let B_i and B_o represent two flat structuring elements with the same shape. Let $S(B)$ represent the size of structuring element B, which is the radial distance from the center of B to the contour of B. Suppose $S(B_i) < S(B_o)$, and the margin structuring element $\Delta B = B_o - B_i$ is the margin region between B_i and B_o. In this paper, B_o represents the structuring element with larger size than target region and is called outer structuring element. B_i represents the structuring element whose size is smaller than $S(B_o)$ and is called

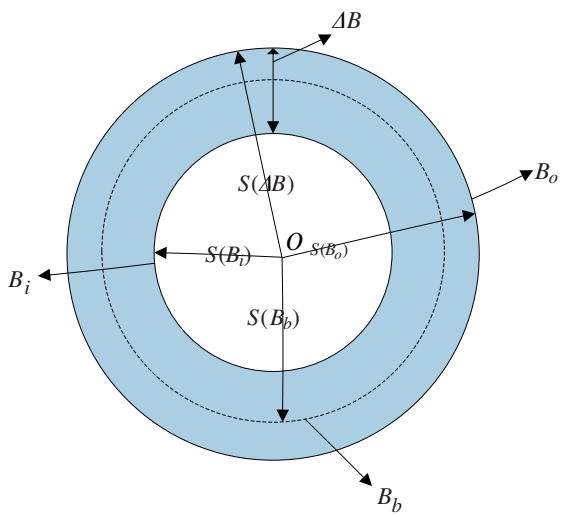

Fig. 1. Relationship of the structuring elements.

inner structuring element. B_b represents the structuring element whose size is between $S(B_o)$ and $S(B_i)$. $S(\Delta B)$ is the radial distance from the center of ΔB to the outer contour of ΔB, then $S(\Delta B)=S(B_o)$. Let $M(\Delta B)$ be the radial distance from the inner contour to the outer contour of ΔB, then $M(\Delta B)=S(B_o)-S(B_i)$. The relationship of B_i, B_o, ΔB and B_b is demonstrated in Fig. 1. In Fig. 1, O is the origin of the structuring elements.

Real target region is different from surrounding background. And, this difference can be utilized by importing the pixels of surrounding regions into top-hat transformations through constructing the structuring elements and re-organizing the operation of top-hat transformation.

Define the operations $f\blacksquare B_{oi}$ and $f\square B_{oi}$ as follows:

$$(f\blacksquare B_{oi})(x,y)=(f\oplus\Delta B)\ominus B_b,$$

$$(f\square B_{oi})(x,y)=(f\ominus\Delta B)\oplus B_b.$$

Then, the new top-hat transformation can be defined as follows:

$$NWTH(x,y)=f(x,y)-f\blacksquare B_{oi}(x,y),$$

$$NBTH(x,y)=f\square B_{oi}(x,y)-f(x,y).$$

B_{oi} represents that the operation is related to B_o and B_i.

The definitions indicate that the new top-hat transformations are different from classical top-hat transformations.

Diff. 1. The sequences of dilation and erosion of new top-hat transformations are different from classical top-hat transformations.

Take *WTH* and *NWTH*, for example, the opening operation in *WTH* is erosion followed by dilation, but the ■ operation in *NWTH* is dilation followed by erosion. In operation ■, $f\blacksquare B_{oi}$ firstly uses $f\oplus\Delta B$ to replace the pixels of the target region with the pixels in the surrounding regions. Then, the following $\ominus$ will only operate the pixels of surrounding regions. This will decrease the number of pixels used for calculation and thus suppress the effect of noises. Simultaneously, firstly operating $\oplus$ operation in *NWTH* will suppress the effect of negative noise in target region because of the maximum operation in dilation, and the following $\ominus$ operation can suppress the effect of positive noises because of the minimum operation in erosion. So, *NWTH* is better than *WTH* on noise suppression and utilizing of the difference information between the target and surrounding regions. *BTH* and *NBTH* have the similar properties.

Diff. 2. New top-hat transformations use two different but correlated structuring elements, and the margin structuring element ΔB is used to utilize the difference information between target and surrounding regions.

Diff. 1 and the definitions indicate that new top-hat transformation firstly replaces the pixels in target region with the pixels of surrounding region through ΔB, and then outputs the difference between target and surrounding regions through the subtraction operation. So, ΔB directly imports the difference information between target and surrounding regions into the structuring elements of new top-hat transformations. And, B_b can change the size of target region used for the calculation of top-hat transformation. Therefore, no matter how small the difference between the target and surrounding regions is, the new top-hat transformation will output this difference in the result image. This is very useful for dim target detection. All of these result in the good utilizing of the difference information in new top-hat transformation, which would improve the performance of top-hat transformation for target detection.

4. Properties

The definitions indicate that new top-hat transformations are different from classical top-hat transformations, and new top-hat transformations form new operations through the two different but correlated structuring elements. Then, new top-hat transformations possess some new properties.

(1) If $M(\Delta B)\neq 0$, new top-hat transformations do not satisfy: $f\geq f\square B_{oi}$ and $f\leq f\blacksquare B_{oi}$.

Classical opening operation $f\circ B$ in *WTH* smoothes bright region of image, then it satisfies: $f\geq f\circ B$. Similarly, there is $f\leq f\bullet B$. But, in $f\square B_{oi}$, the operation sequence of dilation and erosion is different from that of $f\circ B$. Also, $f\square B_{oi}$ uses the margin region of the structuring element (ΔB) to fulfill new top-hat transformation, which means $f\square B_{oi}$ uses the pixels of surrounding regions to replace the pixels of target region. Then, if the processed region is not a target region, the relationship of the pixels in the processed and surrounding regions is not confirmed, which results in the unconfirmation of the relationship of f and $f\square B_{oi}$. Similarly, the relationship of f and $f\blacksquare B_{oi}$ is also not confirmed.

This property indicates that there may be negative values in *NWTH*. To avoid this situation, *NWTH* can be modified as follows:

$$\begin{aligned}NWTH(x,y)&=\max(f(x,y)-f\blacksquare B_{oi}(x,y),0)\\&=\max(f(x,y)-f\blacksquare B_{oi}(x,y),f(x,y)-f(x,y))\\&=f(x,y)-\min(f\blacksquare B_{oi}(x,y),f(x,y)).\end{aligned}$$

Let $NC_{oi}(x,y)=\min(f\blacksquare B_{oi}(x,y),f(x,y))$, then the modified *NWTH* can be defined as follows:

$$MNWTH(x,y)=f(x,y)-NC_{oi}(x,y).$$

Also, the modified *NBTH* can be defined as

$$MNBTH(x,y)=NO_{oi}(x,y)-f(x,y),$$

$$NO_{oi}(x,y)=\max(f\square B_{oi}(x,y),-f(x,y)).$$

(2) Fix ΔB, then the larger the $S(B_b)$ is, the stronger the ability of noise smoothing of $f\square B_{oi}$ or $f\blacksquare B_{oi}$ is.

The ability of noise smoothing of one algorithm is the ability of the algorithm to smooth the un-flat regions of image. In mathematical morphology, operations using structuring element with large size could smooth large region of image, and have strong ability of image smoothing. If ΔB is fixed, the margin region of the structuring element is fixed. Then, following the property of morphological dilation and erosion [1], the larger the $S(B_b)$ is, the stronger the ability of the image smoothing of $f\square B_{oi}$ or $f\blacksquare B_{oi}$ is,

which means the ability of noise smoothing of $f\square B_{oi}$ or $f\blacksquare B_{oi}$ is stronger.

(3) If $B_b \neq B_o$, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ will change the size of the regions in original image.

The definitions of dilation and erosion [1] indicate that dilation increases the size of bright regions of image and decreases the size of dark regions of image following the size of structuring element. Also, erosion decreases the size of bright regions and increases the size of dark regions following the size of structuring element. Then, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ would change the size of bright regions and dark regions following $S(\Delta B)$ and $S(B_b)$. If $B_b \neq B_o$, that is $S(\Delta B) \neq S(B_b)$, then the size increasing and decreasing of image regions are different. So, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ will change the size of the regions in original image.

(4) If $B_b=B_o$, the ability of noise smoothing of $f\square B_{oi}$ and $f\blacksquare B_{oi}$ is strong. And, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ will not change the size of the regions in original image.

If $B_b=B_o$, then $S(B_b)=S(B_o)$ achieves the largest value. So, following property (2), the ability of noise smoothing of $f\square B_{oi}$ and $f\blacksquare B_{oi}$ is strong.

If $B_b=B_o$, then $S(\Delta B)=S(B_b)$, then the size increasing and decreasing of the image regions are the same. So, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ will not change the size of the regions in original image.

(5) Fix B_b, then the larger the $M(\Delta B)$ is, the worse the ability of noise smoothing of $f\square B_{oi}$ or $f\blacksquare B_{oi}$ is.

If B_b is fixed, a larger $M(\Delta B)$ means more pixels used for the calculation of $f\square B_{oi}$ and $f\blacksquare B_{oi}$. Then, following the analysis of Diff. 1, the noise suppression ability of $f\square B_{oi}$ or $f\blacksquare B_{oi}$ becomes worse.

(6) If $B_b=B_o$ and $M(\Delta B)$ achieves the largest value, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ become the classical opening and closing operations.

If $M(\Delta B)$ achieves the largest value, that is $M(\Delta B)=S(B_o)$, then $\Delta B=B_o$. Also, $B_b=B_o$. That is $\Delta B=B_b$, this means $f\square B_{oi}$ and $f\blacksquare B_{oi}$ use two same structuring elements B_o to fulfill the operation. That is

$$f\blacksquare B_{oi}(x,y) = (f \oplus B_o) \ominus B_o = f \bullet B_o(x,y),$$

$$f\square B_{oi}(x,y) = (f \ominus B_o) \oplus B_o = f \circ B_o(x,y),$$

so, $f\square B_{oi}$ and $f\blacksquare B_{oi}$ become the classical opening and closing operations.

5. Multi-scale operations

5.1. Definitions

Multi-scale operation of mathematical morphology could apparently increase the adaptability of mathematical morphological operations for image processing. So, the theory of multi-scale mathematical morphology is widely researched [12,13]. New top-hat transformations are derived from mathematical morphology. Then, the theory of multi-scale operation of new top-hat transformation can also be derived from the basic theory of multi-scale mathematical operation [12,13].

Two different structuring elements, B_b and ΔB, are used to calculate new top-hat transformation. So, the multi-scale operations of new top-hat transformation are different from classical top-hat transformation. Because of B_b and ΔB, there are three types of multi-scale operations.

(1) *The first type of multi-scale operation*: The size of B_b is changeable, but the size of ΔB is determined by B_o and B_i and is un-changeable. So, if B_o and B_i are fixed, ΔB is also fixed, but B_b is changeable. Therefore, the first type of multi-scale operation of new top-hat transformation can be defined as follows.

Suppose the sizes of B_o and B_i are $S(B_o)=m$, $S(B_i)=n$, $m>n$, then $M(\Delta B)=m-n$, $S(\Delta B)=m$, $n \le S(B_b) \le m$. If $S(B_b)$ is changeable, the multi-scale operation can be formulated as follows.

$$(f\blacksquare B_{oi})^c = (f_{c-1} \oplus \Delta B) \ominus B_c, \quad 1 \le c \le m-n;$$

$$(f\square B_{oi})^c = (f_{c-1} \ominus \Delta B) \oplus B_c, \quad 1 \le c \le m-n,$$

where f_{c-1} can be the original image f or the result image of the previous scale or some specified results of f. In these operations, B_b used in $(f\blacksquare B_{oi})^c$ and $(f\square B_{oi})^c$ $(1 \le S(B_b) \le m-n)$ is B_{n+c-1}, and the size is $S(B_{n+c-1})=S(B_i)+c-1$. Based on these multi-scale operations, the multi-scale new top-hat transformation can be formulated in different ways following different application purposes.

This definition only operates the multi-scale operation inside B_o. So, these multi-scale operations only exploit the multi-scale image features in the local area overlaid by B_o. However, it would be more useful for the local feature extraction.

(2) *The second type of multi-scale operation*: The second type of multi-scale operations is based on gradually changing the size of all the structuring elements used in new top-hat transformation. This is the basic idea of multi-scale mathematical morphology [12,13]. The size of ΔB is decided by B_o and B_i. So, if B_b is un-changeable between B_o and B_i. Then,

$$(f\blacksquare B_{oi})_c = (f_{c-1} \oplus \Delta B_c) \ominus B_{bc}, \quad 1 \le c;$$

$$(f\square B_{oi})_c = (f_{c-1} \ominus \Delta B_c) \oplus B_{bc}, \quad 1 \le c,$$

where f_{c-1} can be the original image f or the result image of its previous scale or some specified results of f. In these operations, the sizes of ΔB and B_b used in $(f\blacksquare B_{oi})_r$ or $(f\square B_{oi})_r$ $(1 \le r \le c)$ are gradually increased as $S(\Delta B_r)=S(\Delta B_1)+r-1$, $S(B_{br})=S(B_{b1})+r-1$. These operations also indicates that if ΔB and B_b are confirmed in one certain application, we just need to gradually increase the sizes of ΔB and B_b in multi-scale operations and do not need to reconsider the sizes of B_o and B_i, which decreases the difficulty of utilizing of these multi-scale operations. These multi-scale operations exploit the multi-scale image features in the global area overlaid by B_o. Based on the second type of multi-scale operations, another class of multi-scale new top-hat transformations can be formulated following different application purposes.

(3) *The third type of multi-scale operation*: In the second type of multi-scale operations, if B_b is changeable between B_o and B_i, the first and second type of multi-scale operations can be combined to form the third type of multi-scale operations.

$$(f\blacksquare B_{oi})_{(c-1)(m-n)+d} = (f_{(c-1)(m-n)+d-1} \oplus \Delta B_c) \ominus B_{dc}, \quad 1 \le c,\ 1 \le d \le m-n;$$

$$(f\square B_{oi})_{(c-1)(m-n)+d} = (f_{(c-1)(m-n)+d-1} \ominus \Delta B_c) \oplus B_{dc}, \quad 1 \le c,\ 1 \le d \le m-n,$$

where $f_{(c-1)(m-n)+d-1}$ can be the original image f or the result image of its previous scale or some specified results of f. These multi-scale operations gradually exploit multi-scale image features in the local and global area overlaid by B_o. This would provide more information for the applications of multi-scale new top-hat transformation. Then, based on these multi-scale operations, more powerful multi-scale new top-hat transformations can be formulated following different application purposes.

Moreover, to avoid the negative values in the result of multi-scale operations, the improvement forms can be directly used in the three types of multi-scale operations through replacing $\blacksquare$ and $\square$ operations by using NC_{oi} and NO_{oi}.

In this paper, we want to fully analyze all the aspects of new top-hat transformation and verify the efficient performance on infrared small target detection. So, we give basic definitions of

multi-scale operations in this subsection and indicate the potential powerful ability of image processing through an example of impulsive noise suppression in the next subsection.

5.2. One example for impulsive noise suppression

One image f with impulsive noises can be formulated as

$$f = f_{Ori} + N,$$

where f_{Ori} is the original image without noises, N represents impulsive noises. Then,

$$f_{Ori} = f - N.$$

The gray values of impulsive noise pixels are very large or small, which are very different from the gray values of its surrounding regions. So, new top-hat transformations can be used to identify noise pixels. Here, to avoid negative values in new top-hat transformation, we use *MNWTH* and *MNBTH*. That is, noise pixels N can be identified by *MNWTH* or *MNBTH*. The second type of multi-scale operations is used here to show this example.

If $N(x, y)$ is positive impulsive noise, then the positive noise pixel (x, y) can be identified by $MNWTH(x,y)$. So, the reconstructed image without positive impulsive noise can be expressed as

$$\begin{aligned}\tilde{f}_p(x,y) &= f(x,y) - N(x,y) = f(x,y) - MNWTH(x,y)\\ &= f(x,y) - (f(x,y) - NC_{oi}(x,y)) = NC_{oi}(x,y).\end{aligned}$$

Then, the reconstructed image without positive impulsive noise pixels at scale t is

$$f_p^t = NC_{oi}^t = \min(f_p^{t-1} \blacksquare B_{oi}^t, f_p^{t-1}) = \min((f_p^{t-1} \oplus \Delta B_t) \ominus B_{bt}, f_p^{t-1}).$$

Iteratively,

$$\begin{aligned}f_p^{t-1} = NC_{oi}^{t-1} &= \min(f_p^{t-2} \blacksquare B_{oi}^{t-1}, f_p^{t-2}) = \min((f_p^{t-2} \oplus \Delta B_{t-1})\\ &\ominus B_{bt(t-1)}, f_p^{t-2}),\end{aligned}$$

...

$$f_p^1 = NC_{oi} = \min(f \blacksquare B_{oi}^1, f) = \min((f \oplus \Delta B_1) \ominus B_{b1}, f).$$

Usually, the noises existed in different scales. So, the positive impulsive noises at all scales can be suppressed by using f_p^t at all scales as follows:

$$F_p = f_p^t(f_p^{t-1} \ldots (f_p^1) \ldots).$$

Similarly, the negative noise pixel (x, y) can be identified by $MBWTH(x, y)$. So, the reconstructed image without negative impulsive noise can be expressed as

$$\begin{aligned}\tilde{f}_n(x,y) &= f(x,y) - N(x,y) = f(x,y) + MBWTH(x,y)\\ &= f(x,y) + (NO_{oi}(x,y) - f(x,y)) = NO_{oi}(x,y).\end{aligned}$$

Then, the reconstructed image without negative impulsive noise pixels at scale q is

$$f_n^q = NO_{oi}^q = \max(f_n^{q-1} \square B_{oi}^q, f_n^{q-1}) = \max((f_n^{q-1} \ominus \Delta B_q) \oplus B_{bq}, f_n^{q-1}).$$

Iteratively,

$$\begin{aligned}f_n^{q-1} = NO_{oi}^{q-1} &= \max(f_n^{q-2} \square B_{oi}^{q-1}, f_n^{q-2}) = \max((f_n^{q-2} \ominus \Delta B_{q-1})\\ &\oplus B_{b(q-1)}, f_n^{q-2}),\end{aligned}$$

...

$$f_n^1 = NO_{oi} = \min(f \square B_{oi}^1, f) = \max((f \oplus \Delta B_1) \oplus B_{b1}, f).$$

So, the negative impulsive noises can be suppressed by using f_p^t at all scales as follows:

$$F_n = f_n^q(f_n^{q-1} \ldots (f_n^1) \ldots).$$

Therefore, the reconstructed image without positive or negative noises at all scales can be expressed as

$$\tilde{f}(x,y) = F_n(F_p(x,y)).$$

To verify the efficiency of $\tilde{f}$ for impulsive noise suppression, the Lena images with different percentages of impulsive noises are used. Impulsive noise suppression has been an active research area and some specialised methods have been proposed to suppress the impulsive noises [14,15], some of them are very efficient. In this paper, we want to show that the multi-scale operations from new top-hat transformation can be used for different applications, and thus show the efficiency of new top-hat transformation. $\tilde{f}$ is mathematical morphology based algorithm. So, we compare $\tilde{f}$ with the well known and widely used mathematical morphology based algorithm to show the efficiency of $\tilde{f}$. Morphological alternating sequential filters (ASFs) have been recognized as one class of important and efficient non-linear filters [16–18]. ASFs are iterative operations of opening and closing using structuring elements with the same shape and increasing sizes. Because of the alternating sequences of opening and closing, ASFs can protect the image details better than one single opening or closing. And, the ability of noise suppression of ASFs is good.

In this experiment, to obtain a good ability of noise smoothing following the properties, we set $M(\Delta B)=1$, $B_b=B_o$. The shapes of structuring elements are rectangle. Because the size of impulsive

Fig. 2. Original image and noise image: (a) original Lena mage; (b) with 5% impulsive noise.

Fig. 3. Result of ASF on Fig. 2(b) (from ASF_1, ASF_2, ASF_3 to ASF_4).

Fig. 4. Result of $\tilde{f}$ on Fig. 2(b) (5%).

noise regions is small, t and q are also small. We set $t=q=3$ in this experiment. More noises will results in larger t and q. Fig. 2 is the original image and noise image with 5% impulsive noise. Fig. 3 is the result of ASFs. Fig. 4 is the result of $\tilde{f}$. The results show that ASFs can smooth the noises, but the image details are also smoothed. However, $\tilde{f}$ can efficiently smooth the noises and well protect the image details.

Moreover, heavy noise can be smoothed by using large scale number through $\tilde{f}$. Fig. 5 is the noise image with 10% impulsive noises and the results by using $\tilde{f}$ with different scale numbers. Fig. 5 shows that using large scale number can smooth heavy noise. And, the image details are also well protected. So, the proposed new operations would be very useful for different applications.

To show the efficiency of $\tilde{f}$ on more image data with different and heavy impulsive noises, Table 1 presents the PSNRs of ASFs and $\tilde{f}$ on Lena image with different percentages of impulsive noises. PSNR is defined as follows:

$$\mathrm{PSNR} = 10\log_{10}\frac{\sum_x\sum_y 255^2}{\sum_x\sum_y\{f(x,y)-f_n(x,y)\}^2}.$$

Table 1 presents that because of the efficient performance of new top-hat transformation, $\tilde{f}$ performs better than ASFs. ASFs are efficient filters from the classical top-hat transformation. $\tilde{f}$ is from the new top-hat transformation. These experimental results indicate that the new top-hat transformation would be more efficient than the classical top-hat transformation for different applications.

6. Application for infrared dim small target detection

One of the most important applications of top-hat transformation is infrared dim small target detection. This paper will demonstrate how to use new top-hat transformation to detect infrared dim small target under the conditions of heavy clutter background and noises. *MNWTH* and *MNBTH* not only possess the superiority of new top-hat transformations, but also avoid the negative values in the result image. So, this paper will mainly use *MNWTH* to detect infrared small targets.

New top-hat transformation uses the margin region (ΔB) and inner region (B_b) of structuring elements to fulfill the new operation through changing the operation sequence of dilation and erosion, which uses the difference information between the target and surrounding regions well. Therefore, appropriate selection of the structuring elements following the properties of new top-hat transformation will apparently enhance the ability of top-hat transformation for infrared dim small target detection.

Infrared dim small target is usually a small bright region of image [7–9]. If the target region is bright, the background of the image is not very clutter and the noise of the image is not heavy, the small bright region can be treated as a noise region. In this case, the small target can be detected through filtering the target region as noise region by using the way in [11]. Actually, in some cases, the small target usually existed in the clutter background and the small target image has different noises. Under this situation, the small target region cannot be simply treated as noise region.

Target region in image is usually different from the surrounding background, which is the reason of top-hat transformation being used for target detection. In fact, the definition of new top-hat transformation indicates that new top-hat transformation firstly replaces the pixels in target region with the pixels of surrounding region by ΔB, and then the following operations are all performed on the values imported through ΔB. Therefore, the key of small target detection under the conditions of clutter background and noise image by using the new top-hat transformation is appropriately selecting ΔB.

In the application of infrared small target detection, ΔB represents the surrounding region of the target region. B_b represents the region over the target region. To efficiently detect the infrared small target, the inner size of ΔB should be larger than the size of the possible small targets. In many cases, the size of the small target is usually very small and can be obtained from the prior knowledge. So, it is easy to set the size of ΔB.

Clutter background and noises of the image are also bright pixels in image. They may be the false alarms in the detection result. To suppress the effect of noise and clutter background, the ability of noise smoothing of $f \blacksquare B_{oi}$ should be decreased. So, $S(B_b)$ should be decreased and $M(\Delta B)$ should be increased following the properties (2), (4) and (5).

In this experiment, to be efficient and robust, we set $S(B_b)=S(\Delta B)=21$, $M(\Delta B)=5$ for the experiment in Sections 6.1 and 6.2. Also, the effect of the size changing of $S(B_b)$ and $M(\Delta B)$ for target detection is also discussed in Section 6.3.

Fig. 5. Results on noise images (with 10% impulsive noise) by using different scales: (a) original noise image (10% impulsive noise); (b) result using two scales (PSNR=29.5478); (c) result using three scales (PSNR=32.4618); (d) result using four scales (PSNR=32.7210).

Table 1
Comparison results on Lena image with different percentage of impulsive noises (dB).

PSNR	Filters				
Noise images (%)	ASF_1	ASF_2	ASF_3	ASF_4	$\tilde{f}$
2	25.7564	26.2546	25.7735	26.2252	38.3359
5	25.3701	25.9119	25.4061	25.9908	35.6648
10	24.7989	25.1566	24.7605	25.2660	32.4140
15	23.7050	23.8132	23.7569	24.0371	29.4005
20	22.0391	22.4231	21.9674	22.2381	24.9655
25	19.0051	18.1536	18.9017	18.7012	21.1614
30	14.2196	13.7645	14.5148	13.6019	17.1896

6.1. Target enhancement

Top-hat transformation eliminates the background of infrared small target image through target enhancement, which suppresses the effect of clutter background or noises for target detection [7–10]. So, if top-hat transformation eliminates more clutter background and enhances target better, the target can be detected more easily. To demonstrate the efficient performance of new top-hat transformation for target enhancement, more than 200 infrared dim small target images with different backgrounds and noises are used in this experiment. Some results are shown in Figs. 6–8.

In Figs. 6–8, (a) is the original images with different clutter backgrounds and noises, and the targets are labelled by rectangles. (b) is the enhanced results by using *MNWTH*. (c) is the 3D gray distribution of target and surrounding regions of original images in (a). (d) is the 3D gray distribution of target and surrounding regions in the result images of *MNWTH* demonstrated in (b).

Fig. 6 is an example of small target image with sky and building clutter background. Also, there are some noises existed in the original image (Figs. 6(a) and (c)). But, after *MNWTH*, the clutter background and noises are well suppressed. And, the target region is very bright (Figs. 6(b) and (d)). So, the target can be easily detected.

Fig. 7 is an example of dim small target image with sky and sea clutter background. The target is dim and unclear (Figs. 7(a) and (c)). But, after *MNWTH*, the clutter background is well suppressed and the target region is very clear (Figs. 7(b) and (d)). Then, the dim target can be easily detected.

Fig. 8 is an example of very dim small target image with sky and sea clutter background. Also, after *MNWTH*, the clutter background is suppressed and the target region is very

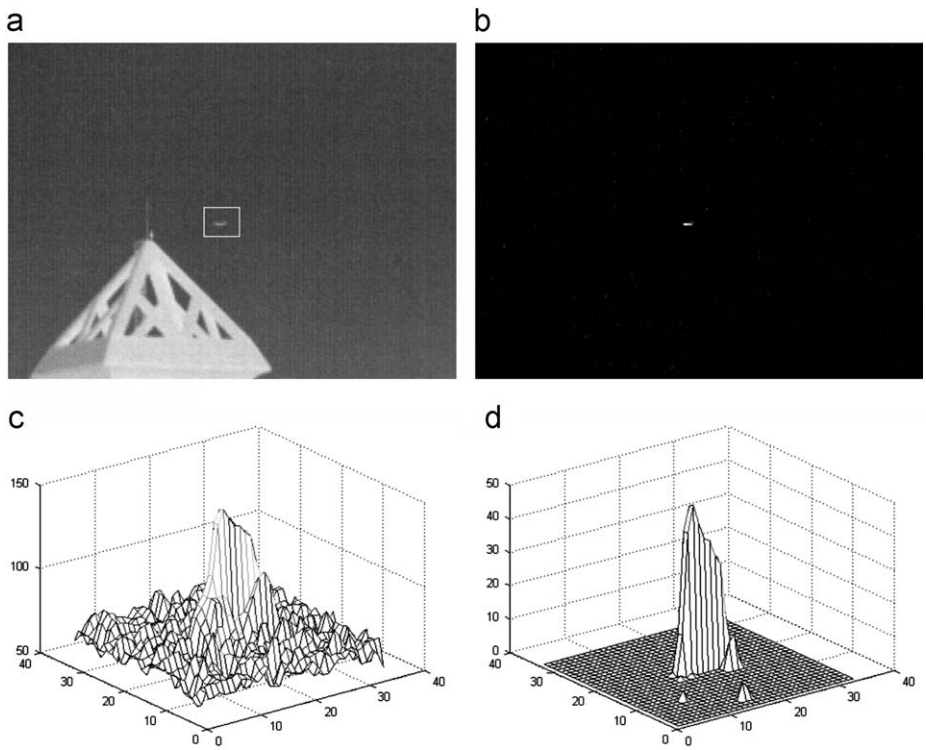

Fig. 6. Infrared dim small target enhancement: (a) infrared dim small target image with sky and building clutter background; (b) enhanced result of (a) by using *MNWTH*; (c) 3D gray distribution of the target and surrounding regions; (d) 3D gray distribution of the target and surrounding regions of the result image of *MNWTH*.

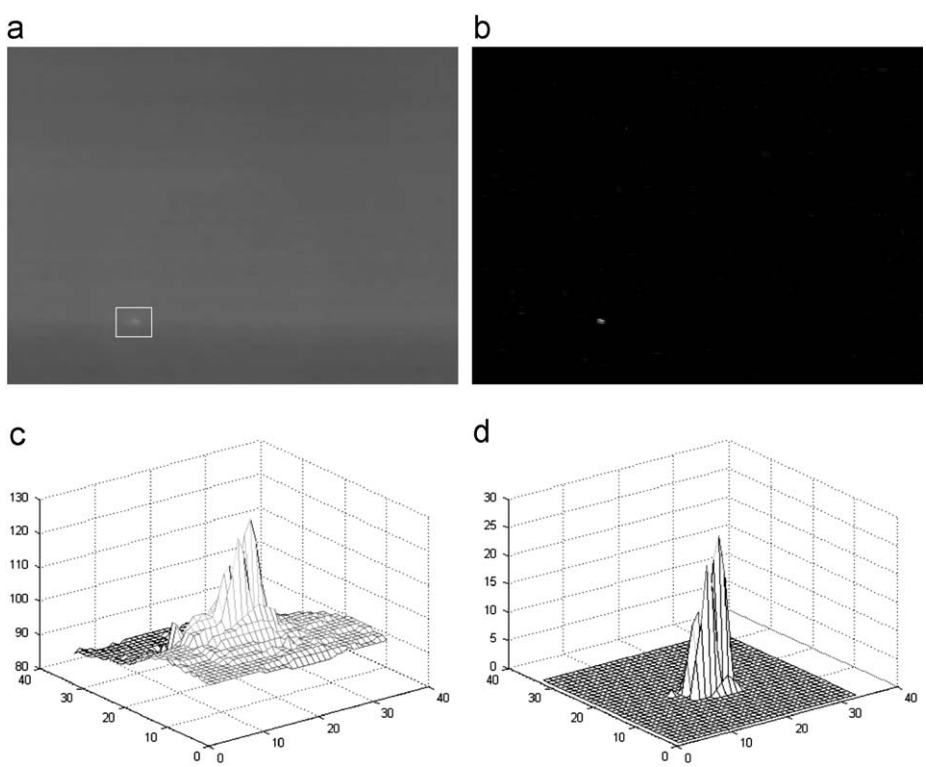

Fig. 7. Infrared dim small target enhancement: (a) infrared dim small target image with sky and sea clutter background; (b) enhanced result of (a) by using *MNWTH*; (c) 3D gray distribution of the target and surrounding regions; (d) 3D gray distribution of the target and surrounding regions of the result image of *MNWTH*.

clear (Figs. 8(b) and (d)). Then, the dim target can be easily detected.

These experimental results demonstrate the efficient performance of new top-hat transformation for small target enhancement.

6.2. Performance comparison

In order to demonstrate the efficient performance of new top-hat transformation for infrared dim small target detection, several widely used methods, including U-kernel [19], E-kernel

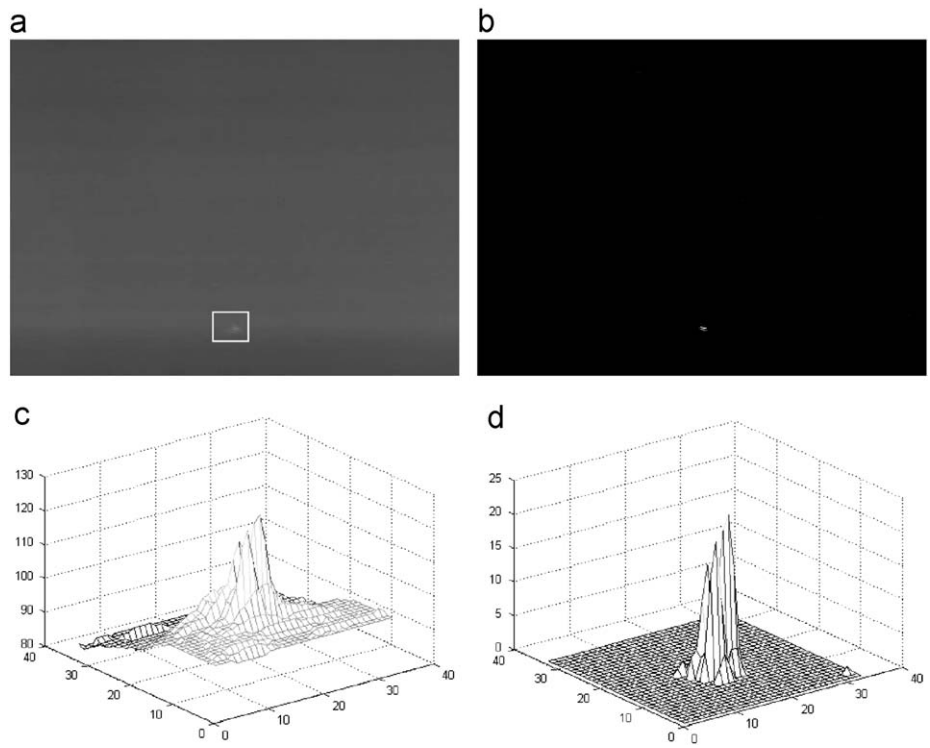

Fig. 8. Infrared dim small target enhancement: (a) infrared dim small target image with sky and sea clutter background; (b) enhanced result of (a) by using *MNWTH*; (c) 3D gray distribution of the target and surrounding regions; (d) 3D gray distribution of the target and surrounding regions of the result image of *MNWTH*.

[19], max-mean [20], max-median [20] and classical top-hat transformation (*WTH*) [7–9], are applied on different infrared small target images to do the comparison. The methods used for comparison are typically efficient and widely used in the field of infrared small target detection. Max-median and max-mean are very efficient for infrared small target detection, and have been widely used. U-kernel and E-kernel are well-known non-parameter methods, and have also been successfully used in different applications. *WTH* is used as the main operation in wide range of applications for small target detection, which is simple and efficient. So, these methods are used for the comparison with our method.

In order to get a good performance, the window size used for max-mean and max-median algorithm has been carefully selected based on the size of the small target. And, we found that 3×3 and 5×5 window size resulted in good performance of max-mean and max-median among most of the images. So, we only show the comparison results of max-mean and max-median with 3×3 and 5×5 window size in the experimental results section.

A measure provided by Soni [11,21] and named local signal-to-background ratio (LSBR) is used in this paper. LSBR is defined as [11,21]

$$\mathrm{LSBR}=10\log\left\{\frac{1}{\sigma_b^2}\sum_{u=-W/2}^{W/2}\sum_{v=-W/2}^{W/2}[I(x-u,y-v)-m_b]^2\right\},$$

where σ_b^2 is the variance and m_b is the mean of the background in the window described by the width and height W around the interest pixel (x, y).

LSBR is an appropriate measure to demonstrate the performance of target detection of various methods [11,21] and is used here for comparison. Small LSBR indicates a dim target with heavy clutter and thus a bad performance of one method for target detection. More than 200 images are used to calculate the LSBR values. Some of the LSBRs after different methods are listed in Table 2.

Not only Table 2 but all the results of the more than 200 images show that the performance of *MNWTH* is better than almost all the other methods on most of the images.

The basic idea of infrared small target detection is efficiently suppressing the clutter background and the noises. Infrared small target region is different from the surrounding clutter background. New top-hat transformation imports the pixels of surrounding clutter background into the calculation, which can efficiently differentiate the small target region from the clutter background and thus efficiently suppresses the clutter background. Moreover, appropriately setting the size of $S(B_b)$ and $M(\Delta B)$ following the properties of new top-hat transformation efficiently suppresses the effect of the noises. So, *MNWTH* could efficiently detect small target and perform better than other widely used methods.

To show the efficient performance of *MNWTH* on more images, the mean value of all the LSBRs of the targets is shown in Fig. 9. Fig. 9 shows that *MNWTH* is more efficient and robust for infrared small target detection than other methods.

This experiment demonstrates the efficient performance of new top-hat transformation for infrared small target enhancement and detection. So, *MNWTH* can well be used to detect the infrared dim small target embedded in images with heavy clutter and noises.

6.3. The effect of size changing of structuring elements

To efficiently suppress the effect of the clutter background and noises, $S(B_b)$ should be decreased and $M(\Delta B)$ should be increased following the properties (2), (4) and (5).

In Fig. 10, the size of ΔB is fixed, which are $S(\Delta B)$=21 and $M(\Delta B)$=5. The size of B_b, $S(B_b)$, is changeable. Corresponding to

Table 2
Comparison of target detection by using LSBR (W=33).

Original image	Max-median (3 × 3) [15]	Max-mean (3 × 3) [15]	Max-median (5 × 5) [15]	Max-mean (5 × 5) [15]	*WTH* [7–9]	U-kernel [14]	E-kernel [14]	MNWTH
0.6267	4.4567	0.9790	0.8528	1.5454	0.5144	2.8753	2.7289	**12.0501**
0.4241	4.6860	0.9003	0.8554	1.2713	0.6033	1.6777	1.6439	**9.2719**
0.6370	6.8404	1.1730	0.7982	1.3755	0.7634	1.2802	1.2374	**8.4764**
0.8522	6.1499	1.2492	0.9679	1.7925	1.1633	1.1675	1.1492	**13.7607**
−0.0862	7.9678	1.7982	1.8357	1.4218	0.2936	0.4216	0.4762	**8.6643**
0.3759	5.0796	0.9617	1.0198	1.3460	0.4680	1.5129	1.5219	**9.6761**
0.3357	4.2591	0.6188	0.4267	0.8903	0.4601	0.9859	0.9403	**5.4243**
0.2766	3.0341	0.6158	0.5510	0.9417	0.4066	2.5067	2.4383	**12.8496**
0.3507	5.1773	0.9277	0.9302	1.0676	0.4175	0.5646	0.6052	**9.4880**
0.1413	4.0689	0.7352	0.4596	0.7349	0.2628	0.3826	0.4054	**7.2664**
0.5900	4.4017	1.1345	1.0189	1.3758	0.7690	0.5905	0.6306	**9.9172**
0.3479	4.0376	1.1506	0.9453	1.2979	0.4853	0.3393	0.3618	**8.6967**
−0.0415	5.9323	1.5636	1.0641	1.2882	0.6816	0.5094	0.5075	**10.9240**
0.3471	3.5088	0.6051	0.6606	1.2090	0.6905	0.6697	0.5873	**9.0456**
0.1044	6.4468	1.4041	1.2786	1.0981	0.3301	0.3904	0.3410	**9.4362**
0.0817	4.6671	1.0675	0.9917	0.9862	0.6640	0.4013	0.4579	**9.0982**
0.7546	4.3146	0.7366	0.5466	1.4560	0.9938	3.7032	4.3396	**5.0929**
0.3290	4.2217	0.8954	0.7482	1.4494	0.9973	7.1693	7.0399	**13.8110**
0.1418	5.1396	0.9709	1.2300	1.1754	0.6348	0.4862	0.4634	**11.9402**
0.2759	4.9350	0.8274	1.0609	1.3344	0.7431	1.2672	1.1444	**15.8468**

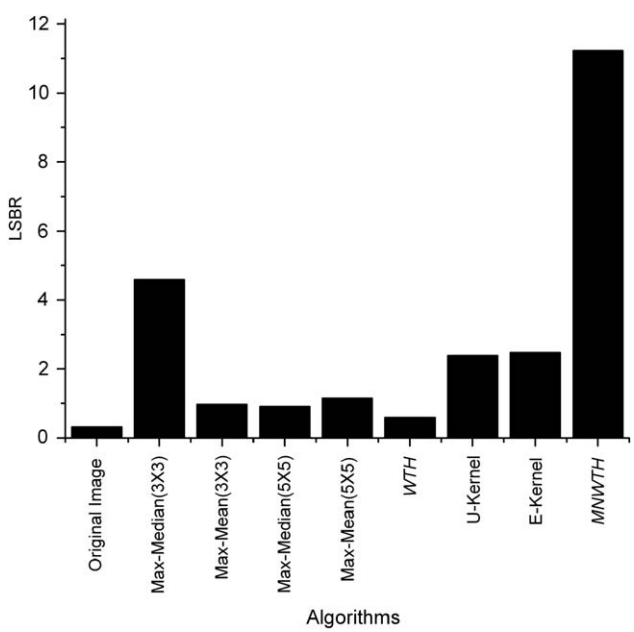

Fig. 9. Mean value of LSBRs of all targets for different algorithms.

one $S(B_b)$, the corresponding LSBR of the target image after *MNWTH* is calculated. The changing of $S(B_b)$ and the corresponding LSBRs are illustrated in Fig. 10. To calculate the LSBR corresponding to one $S(B_b)$ 50 targets have been used. The mean LSBR value of the LSBRs of the 50 targets is used as the LSBR corresponding to one $S(B_b)$. Fig. 10 shows that LSBR decreases following the increasing of $S(B_b)$. So, to efficiently detect the dim target under the conditions of clutter background and noise image, $S(B_b)$ should be small. Actually, $S(B_b)$ should not be too small because the small target usually have a region size. Especially, following the properties (3) and (4), it would be good to set $S(B_b)=S(B_o)=S(\Delta B)$ to keep the size of the image region unchanging. Moreover, experimental results show that setting $S(B_b)=S(B_o)=S(\Delta B)$ is already very efficient for infrared small target detection. So, we set $S(B_b)=S(B_o)=S(\Delta B)$ in this paper.

In Fig. 11, the size of B_b is fixed, which are $S(B_b)$=21. $M(\Delta B)$ is changeable. For one $M(\Delta B)$, the corresponding LSBR of the target image of *MNWTH* is calculated. The changing of $M(\Delta B)$ and the corresponding LSBRs are illustrated in Fig. 11. Also, 50 targets have been used to calculate the LSBR corresponding to one $M(\Delta B)$.

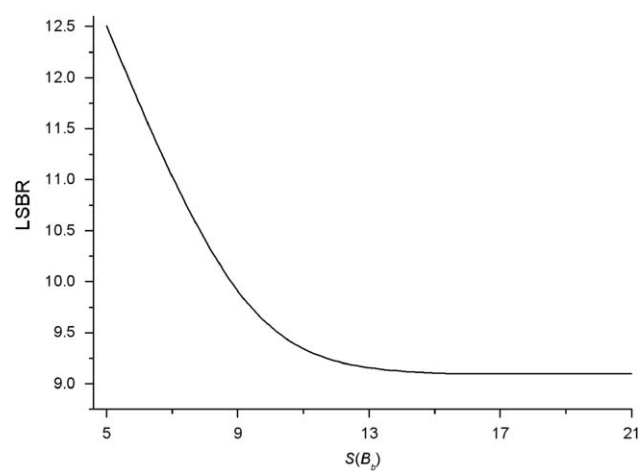

Fig. 10. Effect of size changing of B_b.

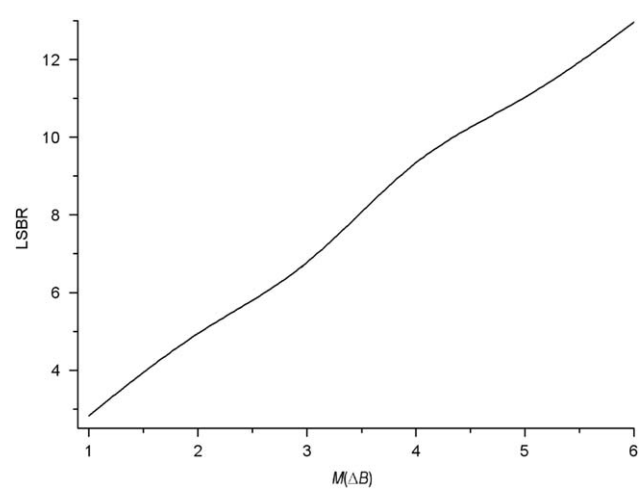

Fig. 11. Effect of size changing of ΔB.

Table 3
Comparison of computational time (s).

Max-median (3×3)	Max-mean (3×3)	Max-median (5×5)	Max-mean (5×5)	WTH	U-kernel	E-kernel	MNWTH
2.290	1.890	2.326	1.948	0.018	0.020	0.022	**0.107**

The mean LSBR value of the LSBRs of the 50 targets is used as the LSBR corresponding to one $M(\Delta B)$. Fig. 11 shows that LSBR increases following the increasing of $M(\Delta B)$. So, to efficiently detect the dim target under the conditions of clutter background and noise image, $M(\Delta B)$ should be large. In fact, $M(\Delta B)$ should not be too large, because $M(\Delta B)$ should ensure that the size of the inner region of ΔB should be larger than the size of the target region in the application of small targe detection. In this paper, we set $M(\Delta B)=5$.

6.4. Computational time comparison

To compare the computational complexity of the methods, some infrared images with size 128×128 are used. The average computational times of different methods are listed in Table 3 (CPU: Intel Pentium 4, 2.6GHz, Memory: 512MB). Table 3 presents that the computational time of *MNWTH* is less than max-mean and max-median, and is very close to other methods, which means the computational complexity of *MNWTH* is low. Moreover, efficient implementation of *MNWTH* could further decrease the computational time of *MNWTH*.

7. Conclusions

Classical top-hat transformation has been widely used in image processing. However, the noises and heavy clutter heavily affect the performance of classical top-hat transformation. Especially, when the classical top-hat transformation is used for infrared dim small target detection, the heavy clutter, noises and dim intensity of target usually make it an ineffective way. The reason is classical top-hat transformation does not well utilize the difference information between the target and surrounding regions.

To overcome the shortcomings of classical top-hat transformation, the new top-hat transformation is studied in this paper. The new top-hat transformation is defined through importing two different but correlated structuring elements and changing the operation sequences of dilation and erosion. Based on the definitions, the properties, multi-scale operations and application for infrared dim small target detection are discussed. The discussions on multi-scale operations and the example for impulsive noise suppression indicate the potential powerful ability of image processing of three types of multi-scale operations. Also, the properties and application for infrared dim small target detection indicate that the importing of the two structuring elements and changing of the operation sequences can well utilize the difference information between the target and surrounding regions, which suppresses the effect of heavy clutter, noises and dim target intensity. Therefore, the new top-hat transformation with the parameters selected following the properties can be well used for infrared dim small target detection. Various experiments verified that new top-hat transformation was better than classical top-hat transformation and other widely used methods for infrared dim small target detection. More importantly, new top-hat transformation and its multi-scale operations can also be used for other image processing applications in wide fields.

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